



PIMS PRO

Pipeline Integrity Management System

Pipeline Corrosion Assessment & Remaining Life Calculator

Version	1.0
Developer	M. Fahad Zafar
Report File	PL-001_Inspection_Report_2026-07-15.pdf
Generation Date	15 July 2026

Inspection Information

Pipe ID	PL-001
Pipeline Name	Crude Oil Transfer Line
Material	API 5L X52
Fluid	Crude Oil
Inspector	M. Fahad Zafar
Inspection Date	15 July 2026
Operating Pressure	5.5 MPa

Engineering Calculations

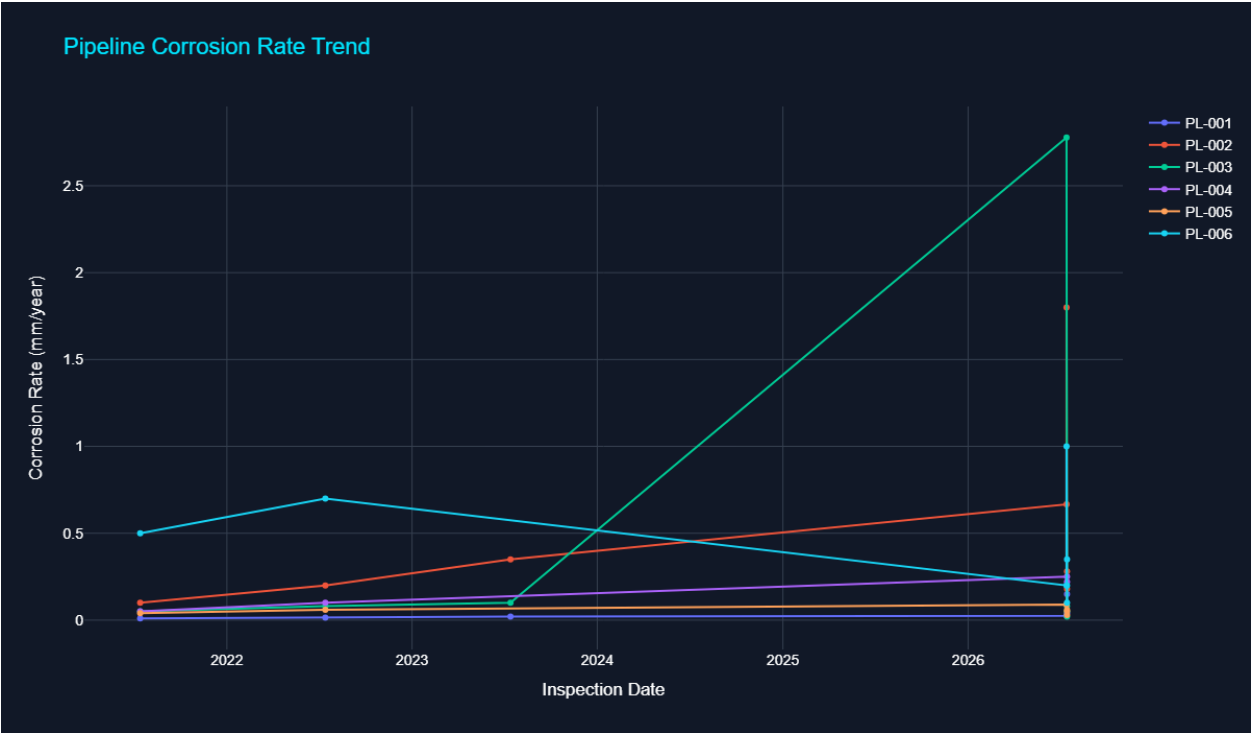
Parameter	Value
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Corrosion Rate	0.15 mm/year
Minimum Thickness	8.34 mm
Remaining Life	22.10 years
Safety Factor	1.42
Corrosion Loss	5.60 %
Health Score	91.35 %
Failure Year	2048

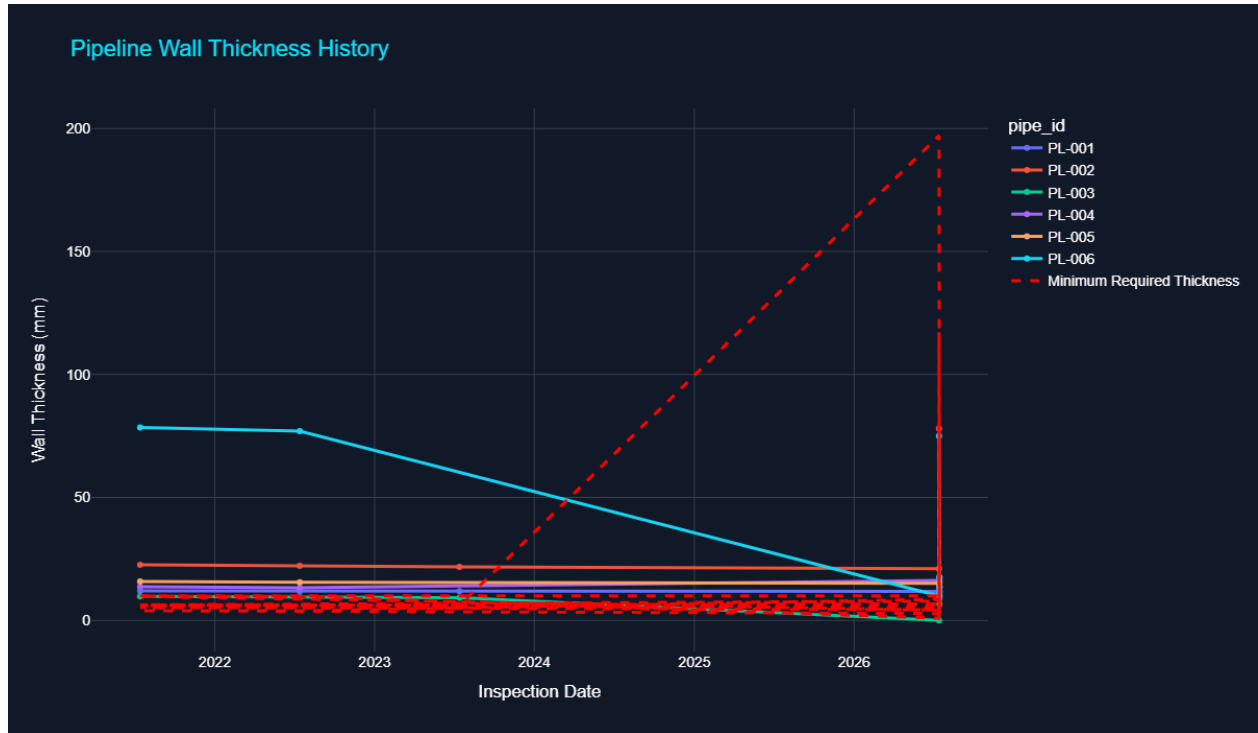
KPI Summary

Health Score	91.35%
Risk Level	Medium
Remaining Life	22.10 Years
Corrosion Rate	0.15 mm/year

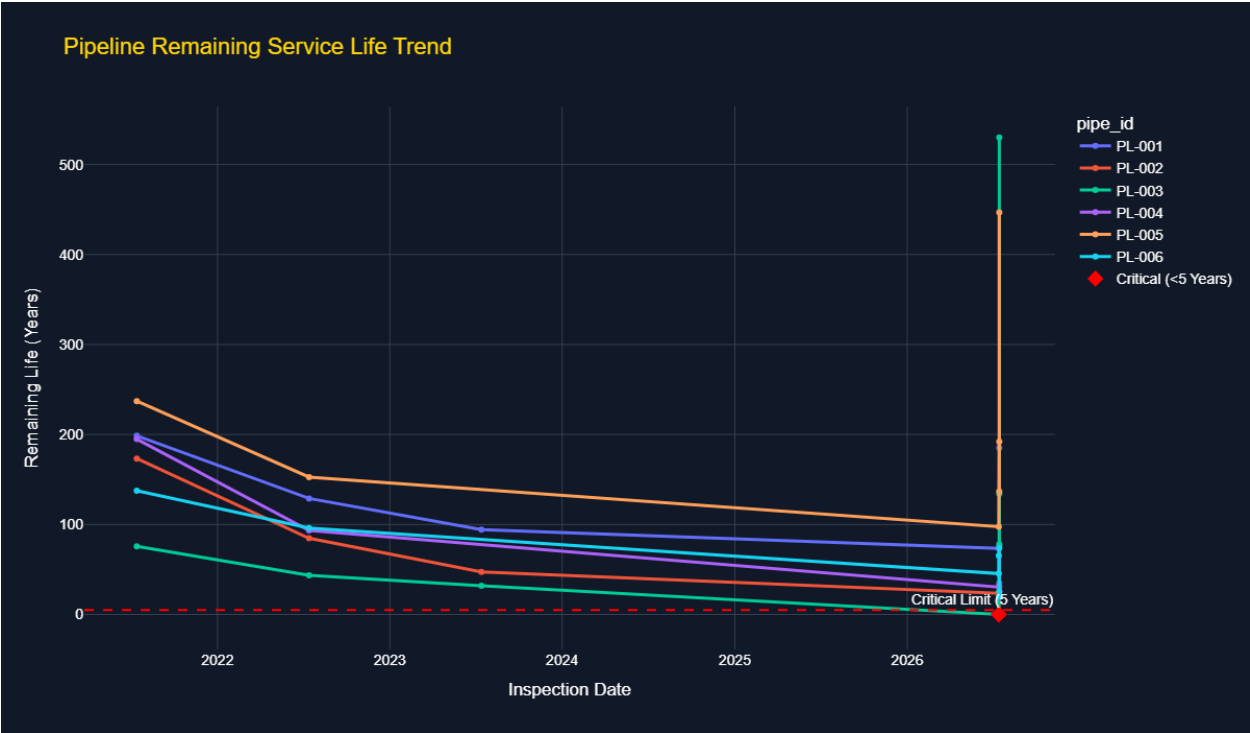
Corrosion Trend



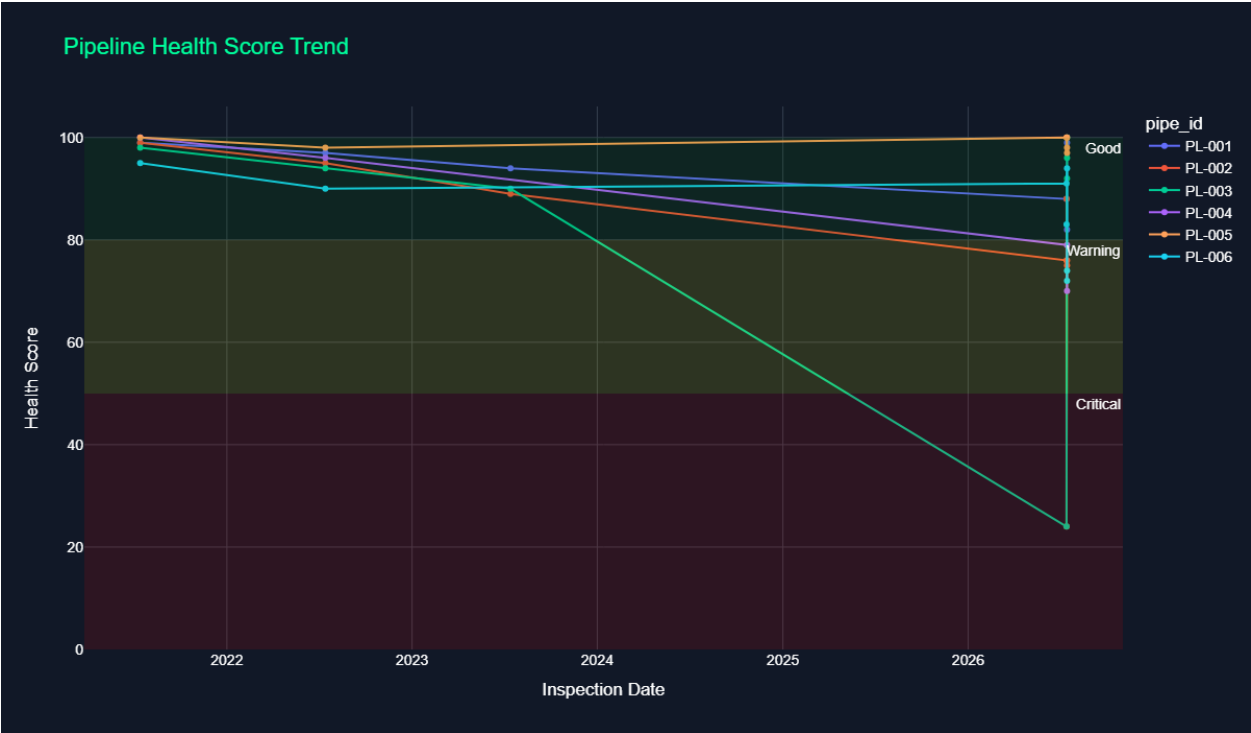
Thickness History



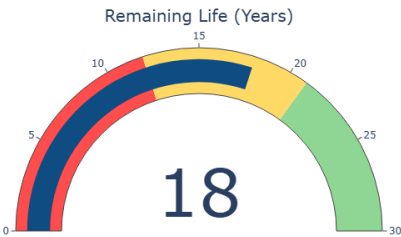
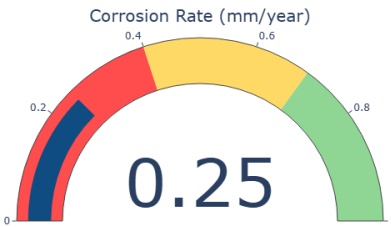
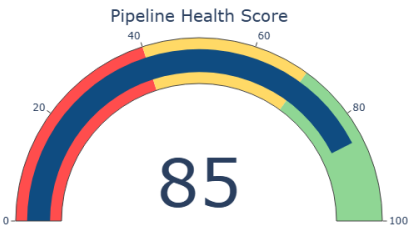
Remaining Life Trend



Health Trend



Engineering KPI Gauges

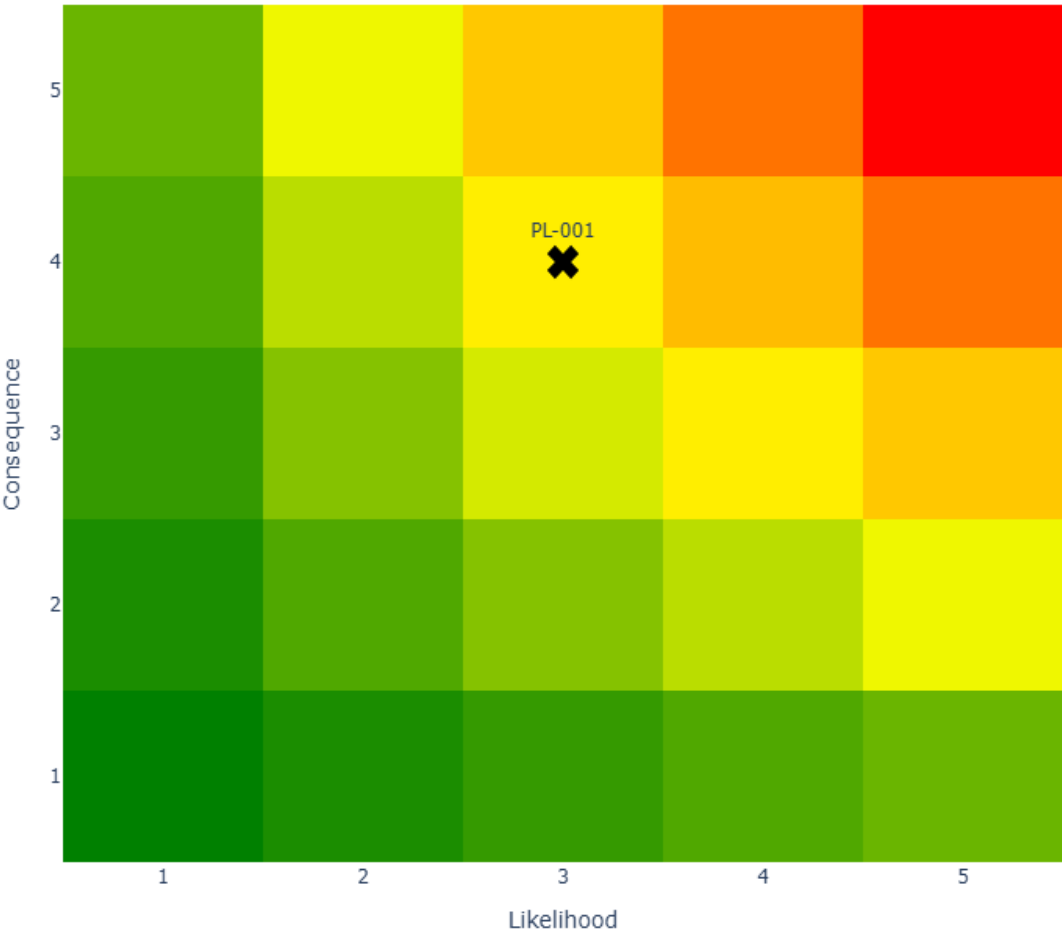


Risk Assessment

Parameter	Value
Likelihood	2
Consequence	3
Risk Score	6
Risk Category	Medium

5 × 5 Risk Matrix

5 × 5 Pipeline Risk Matrix



Engineering Recommendation

Overall Decision	Pipeline Suitable for Continued Service
Maintenance Suggestion	Routine inspection only.
Inspection Interval	Next inspection after 5 years.
Priority	LOW
Warning Message	No immediate engineering concern.

Inspection Summary

The pipeline (PL-001) remains suitable for continued operation. The measured corrosion rate is 0.15 mm/year with an estimated remaining service life of 22.10 years. The calculated health score is 91.3% and the overall risk category is Medium. Routine inspection only.

Conclusion:

Based on the inspection data and engineering calculations performed in accordance with ASME B31.3 and API 570 principles, the pipeline integrity has been evaluated and appropriate maintenance recommendations have been provided to support safe and reliable operation.

Engineering Notes

Engineer observations, maintenance remarks, or site comments may be recorded below.

Engineering Standards

- ASME B31.3 – Process Piping

- API 570 – Piping Inspection Code

The engineering calculations, integrity assessment, corrosion evaluation, remaining life estimation, and inspection recommendations presented in this report are based on the principles and methodologies defined in ASME B31.3 and API 570. The report is intended as an engineering decision-support document and should be reviewed by qualified personnel before maintenance or operational decisions are made.